

Sustained Abstinence in severe Ketamine Use Disorder following Ibogaine Treatment

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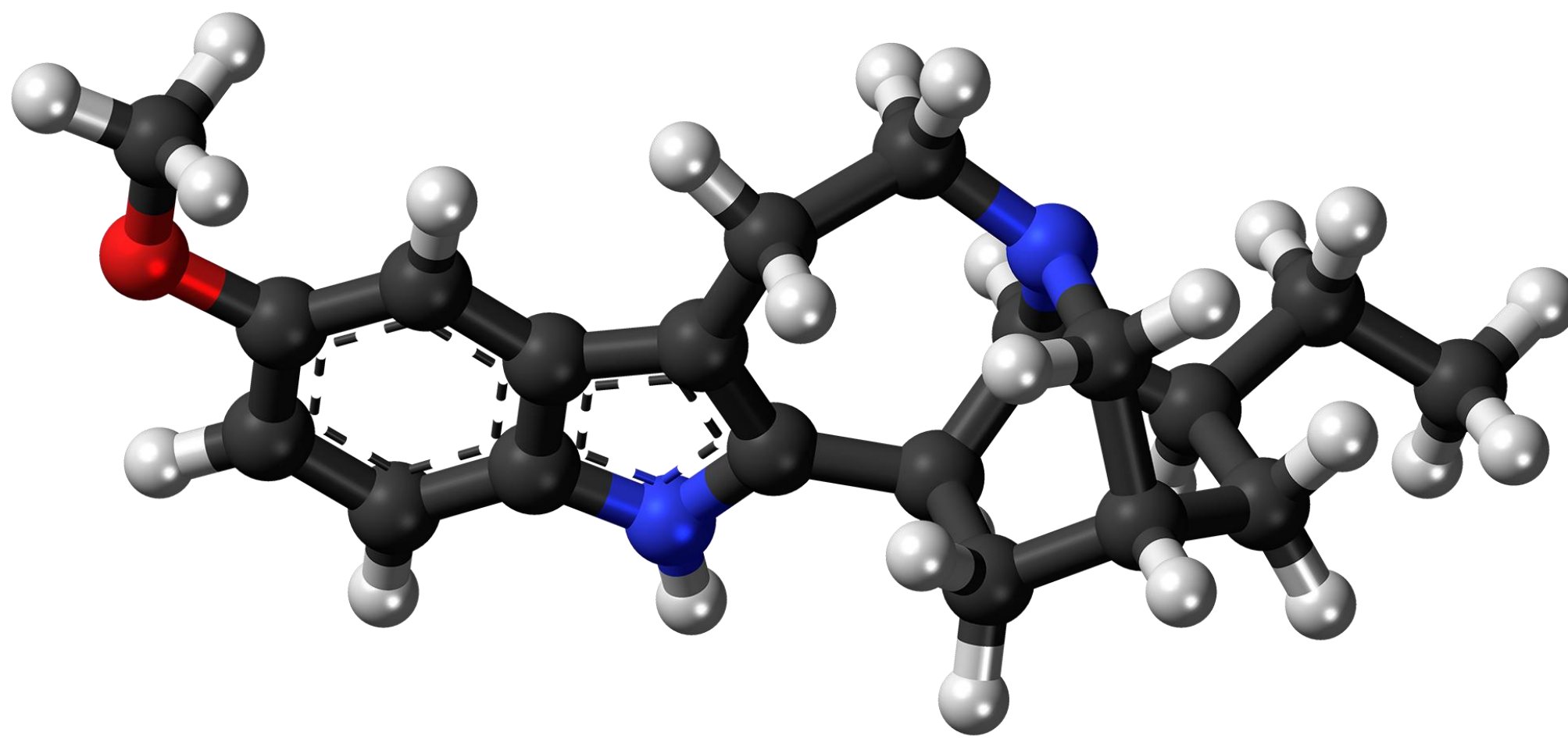
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INTRODUCTION & BACKGROUND

Substance use disorders (SUD) involving ketamine, cocaine, and alcohol present significant clinical challenges, frequently marked by high relapse rates and a lack of approved pharmacological options.

Ibogaine, an indole alkaloid derived from *Tabernanthe iboga*, has demonstrated potential anti-addictive properties but remains unapproved in Europe and North America due to safety risks, including QTc prolongation and potential cardiotoxicity.

This report documents the first longitudinally tracked case of sustained abstinence in severe ketamine use disorder following ibogaine therapy, supported by serial toxicology and standardized psychometric outcomes.



PROPOSED PHARMACOLOGICAL MECHANISMS

We hypothesize a triphasic recovery process driven by ibogaine's unique polypharmacology.

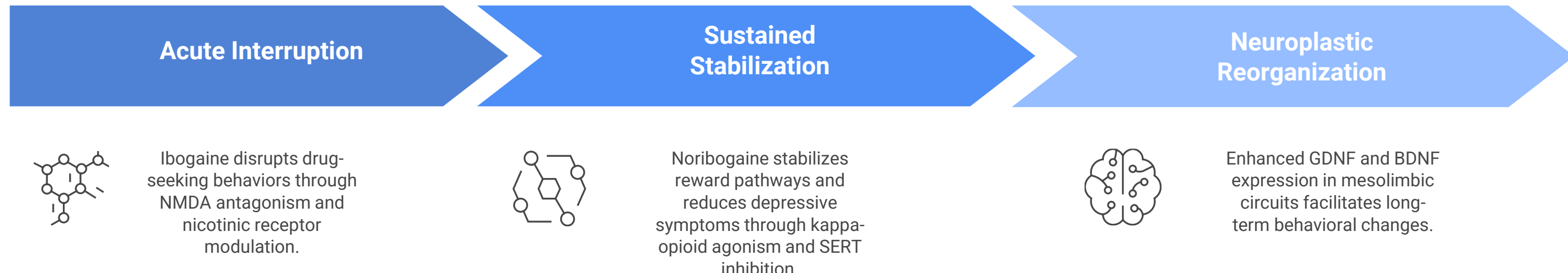


Table 1: Pharmacological Mechanism

Target / Mechanism	Primary Process Affected	Ibogaine Binding	Noribogaine Binding	Mechanistic Notes
NMDA (PCP site)	↓ Phasic dopamine spikes	Moderate antagonist	Weak	Reduces VTA burst firing; interrupts drug-seeking (Govender et al., 2024)
KOR	↓ dopamine responsiveness	Weak	Moderate	Critical anti-addictive and anti-dysphoric effects; KOR agonism is a preserved target in safer analogues (Mash 2024; Govender et al., 2024)
DAT	↑ Tonic dopamine tone	Moderate	Moderate	Stabilizes dopamine baseline (Bulling et al., 2012)
α3β4 nAChR	↓ Phasic dopamine spikes	Strong antagonist	Weak	Weakens conditioned cue linkages; reduces reward & craving (Mash 2024)
σ-1 receptor	Plasticity modulation	Moderate	Weak	Possible TRKB facilitator
SERT	modulates mood and anxiety	Moderate	Strong inhibitor	Main serotonin transporter effect; reuptake blocker (Bulling et al., 2012)
TRKB (via BDNF)	↑ Neuroplasticity	Indirect	Indirect	Downstream convergent pathway (Palhas et al., 2025; Marton et al., 2019; Nardou 2023)
NET	Minimal	Weak	Weak	Not central
5-HT2C	Minor modulation	Weak	Weak	Secondary
MOR	None	Very weak	Very weak	Not mechanistic (Mash 2024; Govender et al., 2024)
DOR	None	Very weak	Very weak	Not relevant (Mash 2024; Govender et al., 2024)
5-HT2A	Minimal psychotropic	Very weak	Very weak	Main target for serotonergic psychedelics (Palhas et al., 2025; Villalba et al., 2024)

Table 1: A systematic comparison of the known binding affinities and proposed functional consequences of ibogaine and its primary active metabolite, noribogaine, across central nervous system (CNS) receptors and transporters implicated in the pathophysiology of substance use disorders. The therapeutic efficacy of ibogaine is hypothesized to arise from a biphasic mechanism: rapid and acute interruption of addiction circuits by ibogaine (primarily via NMDA and nAChR antagonism) followed by prolonged, sustained anti-craving and neuroplastic effects mediated by Noribogaine (predominantly via KOR agonism and SERT inhibition). Data are compiled from in vitro binding assays and in vivo functional studies. N-methyl-D-aspartate (NMDA) receptor, α3β4 nicotinic Acetylcholine Receptor (α3β4 nAChR), Ventral Tegmental Area (VTA), Sigma-1 (Sigma-1) receptor, Serotonin Transporter (SERT), Dopamine Transporter (DAT), Tropomyosin Receptor Kinase B (TRKB), Brain-Derived Neurotrophic Factor (BDNF), kappa-Opioid Receptor (KOR), Norepinephrine Transporter (NET), mu-Opioid Receptor (MOR), delta-Opioid Receptor (DOR), and 5-Hydroxytryptamine (5-HT2A/2C) receptor.

Figure 2: Patient Journey

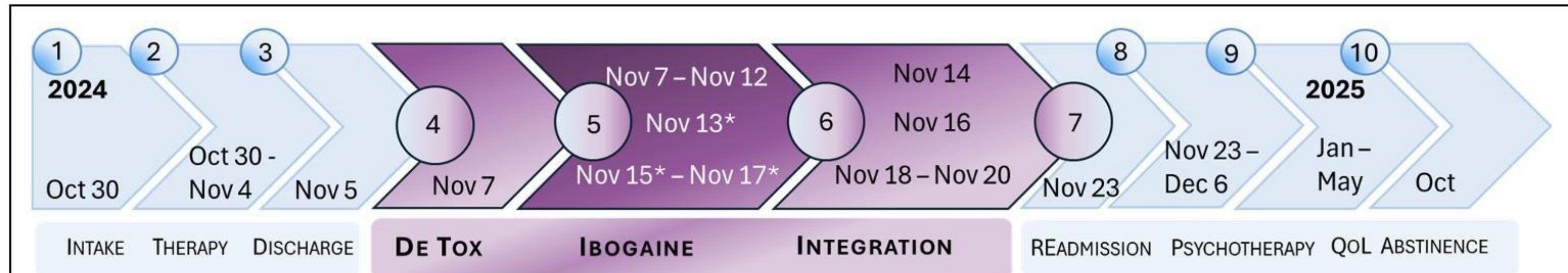


Figure 2: Longitudinal Clinical Trajectory and Treatment Timeline. The schematic illustrates the patient's progression through ten distinct clinical stages: (1) initial presentation at a German psychiatric clinic with severe polysubstance use disorder (ketamine, cocaine, alcohol) and severe depression (BDI-II: 32); (2) receipt of supportive therapy and nutritional stabilization without antidepressants or opioid agonists; (3) discharge against medical advice to seek ibogaine therapy abroad; (4) intake at a 13-day residential program in Mexico including medical screening and a 7-day supervised detox; (5) administration of a 900 mg ibogaine HCl flood dose (13–15 mg/kg) under continuous ECG and 30-minute vital sign monitoring, followed by two 500 mg booster doses; (6) a post-dosing period focused on psychological processing and emotional regulation; (7) medical discharge following the reported complete cessation of cravings for all substances; (8) re-admission to the German clinic for integration therapy with abstinence confirmed via negative toxicology; (9) participation in daily psychotherapy and structured relapse prevention programs; and (10) ten-month longitudinal monitoring via Outcome MD confirming sustained abstinence, absent cravings, minimal depressive symptoms (PHQ-9: 0–3), and significant quality-of-life improvements (WHOQOL-BREF increase from 55 to 71).

CLINICAL OUTCOMES

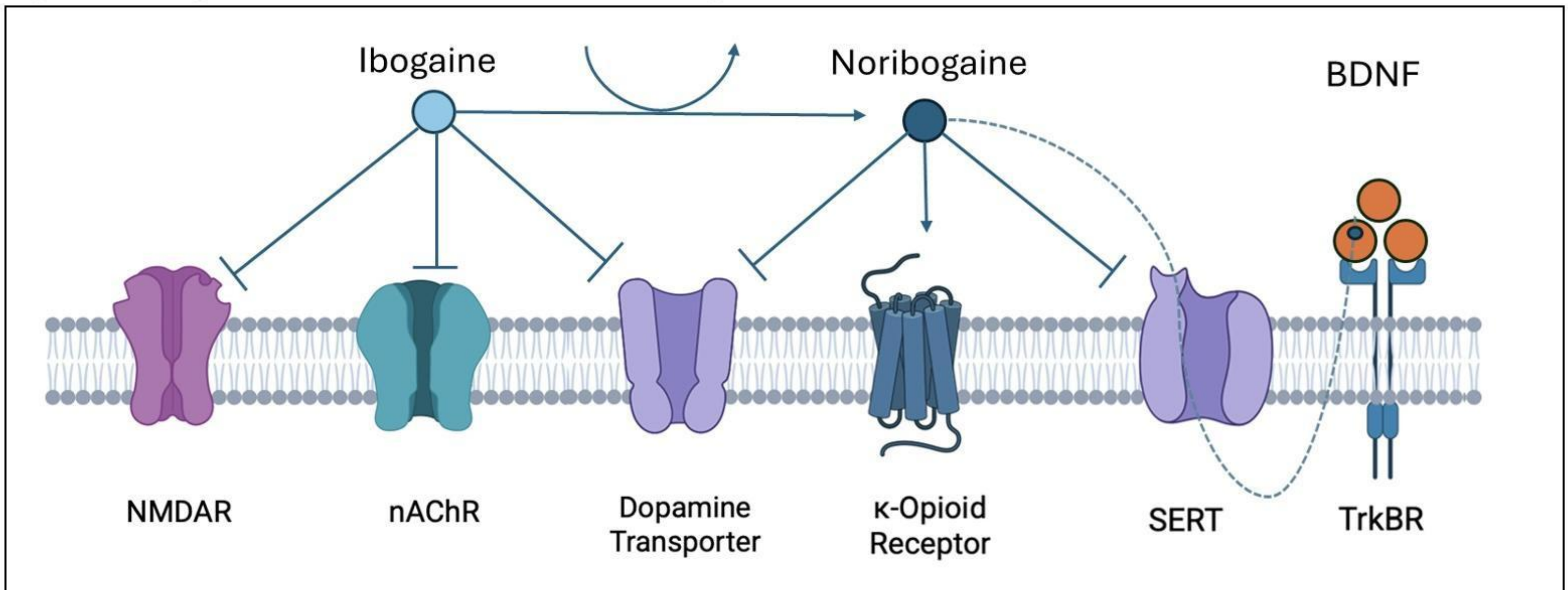
Immediate Efficacy: The patient reported a complete cessation of cravings for ketamine, cocaine, alcohol, nicotine, cannabis, and caffeine within 3 to 5 days post-treatment.
Sustained Abstinence: Over a 10-month longitudinal follow-up, serial toxicology assessments repeatedly confirmed abstinence from all previously misused substances.

Table 2: Longitudinal Psychometric and Clinical Outcomes

A: Mood, Anxiety, and Trauma				
Timepoint	Setting	Depression (BDI-II / PHQ-9)	Anxiety (GAD-7)	PTSD (PCL-5)
Pre-Ibogaine (30 Oct 2024)	German Clinic	BDI-II: 32 (Severe)	Not recorded	Not recorded
Immediately Pre-Ibogaine (7 Nov 2024)	Beond Mexico	PHQ-9: ~18–20 (Moderately severe)	GAD-7: Moderate–Severe (~15+)	Documented trauma exposure, no PCL-5
Post-Ibogaine (23 Nov 2024)	German Clinic	BDI-II: 14 (Mild)	Anxiety: low-moderate	Not measured
Early Follow-up (Jan 2025)	Beond Mexico	PHQ-9: 0–2 (Minimal)	GAD-7: 0–3 (Minimal)	PCL-5: ~55 (improving from ~90 pre-treatment)
Mid Follow-up (May 2025)	Beond Mexico	PHQ-9: 0–3	GAD-7: 0–4	PCL-5: ~52–55
Latest (Oct 2025)	Beond Mexico	PHQ-9: 1–4	GAD-7: 3	No PTSD symptoms causing impairment
B. Craving and Substance Use				
Timepoint	Instrument	Craving Score	Substance Use: Notes	
Pre-Ibogaine (7 Nov 2024)	DUDIT	34 (Severe)	Ketamine daily, cocaine weekly, alcohol frequent High dependence	
	CAGE	4 (Positive for alcoholism)		
	Substance Craving (self-report)	High	Active use	
Post-Ibogaine (23 Nov 2024)	Clinical Observation	0 No Craving	Abstinent (confirmed by urine tox); Patient reports cessation of desire for all substances (including caffeine + nicotine)	
Jan-25	Brief Substance Craving Scale	0	None in past 7 days: Frequency: Never	
May-25	Brief Substance Craving Scale	0	None	
Oct-25	DSM-5 Level 2 — Substance Use	0	None: Sustained remission	
C. Functional Status and Quality of Life				
Timepoint	Parameter	Score	Interpretation	
07 Nov 2024 (pre-Ibogaine)	WHOQOL-BREF Total	55/100	Low QoL	
	Health Satisfaction	3 (Neutral)	Health neither good nor poor	
10 Oct 2025 (post-Ibogaine)	WHOQOL-BREF Total	71/100	Moderate–High QoL	
	Health Satisfaction	5 (Very satisfied)	Significant improvement	
	Daily functioning	Improved	From “dissatisfied” → “satisfied”	
	Work ability	3 → 5 (neutral → very satisfied)		
	Enjoyment of life	3 → 4	Moderate → high enjoyment	
	Sense of meaning	Stable high (4→4)	High meaning maintained	
Longitudinal Psychometric and Clinical Outcomes for the Patient Pre-Ibogaine, Post-Ibogaine, and through 10-Month Follow-up (October 2025). The data are presented in a stacked format to reflect changes across Mood, Craving, and Functional Status.				

References: Perez-Rosal et al. (In Review) Sustained Abstinence in severe Ketamine Use Disorder following Ibogaine Treatment Bulling, S., et al. (2012). The mechanistic basis for noncompetitive ibogaine inhibition of serotonin and dopamine transporters. The Journal of Biological Chemistry, 287(22), 18524–18534. He, D. Y., et al. (2005). Glial cell line-derived neurotrophic factor mediates the desirable actions of the anti-addiction drug ibogaine against alcohol consumption. The Journal of Neuroscience, 25(3), 619–628. Litjens, R. P., & Brunt, T. M. (2016). How toxic is ibogaine?. Clinical Toxicology, 54(4), 297–302. Mash D. C. (2023). IUPHAR - invited review - Ibogaine - A legacy within the current renaissance of psychedelic therapy. Pharmacological Research, 190, 106620. Morgan, C. J., Curran, H. V., & Independent Scientific Committee on Drugs. (2012). Ketamine use: a review. Addiction, 107(1), 27–38. Ona, G., et al. (2023). Main targets of ibogaine and noribogaine associated with its putative anti-addictive effects: A mechanistic overview. Journal of Psychopharmacology, 37(12), 1190–1200.

Figure 1: Major Mechanisms of Action for Ibogaine



1. Acute Dampening of the Reward Signal and Anti-Dysphoria: Ibogaine and its metabolite, Noribogaine, directly reduce the acute, high-amplitude release of dopamine (the primary reinforcement signal) triggered by drugs or associated cues, particularly via NMDA antagonism. This effect is complemented by Noribogaine's critical kappa opioid receptor KOR agonism, which may contribute to the rapid interruption of withdrawal-related dysphoria and craving (Ona et al., 2023; Palhas et al., 2025). This collective action rapidly dampens the exaggerated reward signal and immediately weakens the compulsive drive for use.

2. Synaptic Stabilization and Withdrawal Mitigation: The longer-acting metabolite, Noribogaine, gently modulates the system. Unlike addictive stimulants, Noribogaine acts as a moderate blocker of the dopamine transporter DAT, which clears dopamine from the synapse. This helps maintain a stable, non-peaking level of background dopamine, which prevents the severe emotional crash and dysphoria of withdrawal while simultaneously avoiding the strong, new reward signals that drive addiction (Wells et al., 1999).

3. Disruption of Learned Associations: Ibogaine and Noribogaine chemically disrupt the learned associations that drive addiction. By strongly blocking receptors (specifically the α3β4 nAChR), they weaken the brain's link between environmental cues (triggers) and the automatic, dopamine-driven habit. Individuals report that the triggers simply lose their power to provoke craving (Ivan et al., 2023).

4. Induction of Neuroplasticity: Ibogaine induces a temporary surge in neuroplasticity (the brain's ability to reorganize). By activating pathways such as those involving Brain-Derived Neurotrophic Factor BDNF and its receptor TRKB, it opens a critical window of opportunity where the brain can easily form new, healthier behavioral patterns to replace the previously weakened addiction habits (Marton et al., 2019; Nardou et al., 2023).

ETHICAL CONSIDERATIONS & CONCLUSIONS

Clinical Significance: This case provides rare, rigorously documented evidence supporting ibogaine's potential in treating chronic, complex ketamine dependence that is resistant to conventional therapies.
Ethical Dilemma: The necessity for the patient to seek unapproved care internationally highlights the conflict of integrating a therapy with known mortality risks back into formal psychiatric care.
Future Directions: These findings underscore the urgent need for controlled clinical trials within regulated frameworks to further investigate ibogaine's safety and efficacy.